

Assignment: Build a Star Schema in MySQL

Total Marks: 10

Goal

Design and implement a simple Star Schema in MySQL for Sales and Customer Analytics.

You will create dimension and fact tables, insert sample data, and run basic analytical queries.

Scenario

Your organization wants to analyze sales performance across:

- Customers
- Products
- Time

You need to build a Sales Data Mart using a Star Schema.

Task 1: Design the Star Schema

Create a simple ERD, either on paper or digitally, using the following tables.

dim_customer

- customer_key – Primary Key
- customer_name
- region
- segment

dim_product

- product_key – Primary Key
- product_name
- category
- sub_category

dim_date

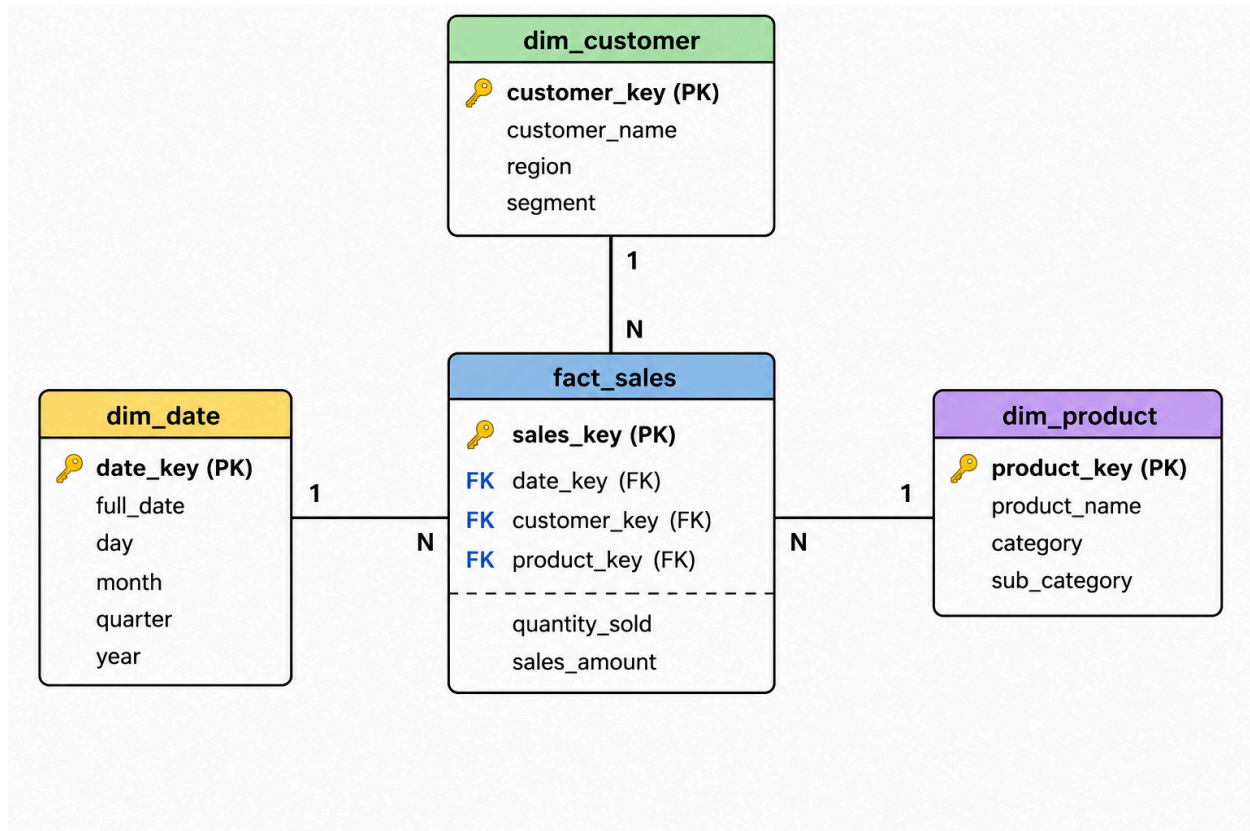
- date_key – Primary Key, in YYYYMMDD format
- full_date
- day
- month
- quarter
- year

fact_sales

- sales_key – Primary Key
- date_key – Foreign Key
- customer_key – Foreign Key

- product_key – Foreign Key
- quantity_sold
- sales_amount

Solution:



Make sure the fact table is connected correctly to all three dimension tables.

Task 2: Create the Tables in MySQL

Create the four tables using SQL.

```

CREATE TABLE dim_customer (
  customer_key INT PRIMARY KEY,
  customer_name VARCHAR(100),
  region VARCHAR(50),
  segment VARCHAR(50)
);

CREATE TABLE dim_product (
  product_key INT PRIMARY KEY,
  product_name VARCHAR(100),
  category VARCHAR(50),
  sub_category VARCHAR(50)
);

CREATE TABLE dim_date (
  date_key INT PRIMARY KEY,

```

```

    full_date DATE,
    day INT,
    month INT,
    quarter INT,
    year INT
);
CREATE TABLE fact_sales (
    sales_key INT PRIMARY KEY,
    date_key INT,
    customer_key INT,
    product_key INT,
    quantity_sold INT,
    sales_amount DECIMAL(10,2),
    FOREIGN KEY (date_key) REFERENCES dim_date(date_key),
    FOREIGN KEY (customer_key) REFERENCES dim_customer(customer_key),
    FOREIGN KEY (product_key) REFERENCES dim_product(product_key)
);

```

Task 3: Insert Sample Data

Insert the following sample records.

dim_customer

```

INSERT INTO dim_customer VALUES
(1, 'Alice Corp', 'North', 'Enterprise'),
(2, 'Beta LLC', 'South', 'SMB');

```

dim_product

```

INSERT INTO dim_product VALUES
(1, 'Laptop Pro', 'Electronics', 'Computers'),
(2, 'Office Chair', 'Furniture', 'Chairs');

```

dim_date

```

INSERT INTO dim_date VALUES
(20250101, '2025-01-01', 1, 1, 1, 2025),
(20250102, '2025-01-02', 2, 1, 1, 2025);

```

fact_sales

```

INSERT INTO fact_sales VALUES
(1, 20250101, 1, 1, 10, 15000.00),
(2, 20250102, 2, 2, 5, 1000.00);

```

Task 4: Run Analytical Queries

Run the following queries.

A. Total Sales Amount

```

SELECT SUM(sales_amount) AS total_sales
FROM fact_sales;

```

B. Sales by Customer

```

SELECT c.customer_name, SUM(f.sales_amount) AS total_sales
FROM fact_sales f
JOIN dim_customer c

```

```
ON f.customer_key = c.customer_key  
GROUP BY c.customer_name;
```

C. Sales by Product Category

```
SELECT p.category, SUM(f.sales_amount) AS total_sales  
FROM fact_sales f  
JOIN dim_product p  
ON f.product_key = p.product_key  
GROUP BY p.category;
```

D. Daily Sales Trend

```
SELECT d.full_date, SUM(f.sales_amount) AS daily_sales  
FROM fact_sales f  
JOIN dim_date d  
ON f.date_key = d.date_key  
GROUP BY d.full_date  
ORDER BY d.full_date;
```

Submission Requirements

Submit exactly 4 clear screenshots:

1. Star Schema ERD – Show the fact table, dimension tables, and their relationships.
2. Created Tables – Show the four tables successfully created in MySQL.
3. Inserted Data – Show the sample data available in the dimension and fact tables.
4. Query Results – Show the results of all four analytical queries.

Also submit your SQL script containing:

- Table creation
- Data insertion
- Analytical queries

Evaluation Rubric – 10 Marks

1. Star Schema Design – 2 Marks

The ERD correctly shows fact_sales, the three dimension tables, and their relationships.

2. Table Creation – 2 Marks

All four tables are created correctly with the required primary and foreign keys.

3. Sample Data – 2 Marks

The provided sample data is inserted correctly into all tables.

4. Analytical Queries – 3 Marks

The four required analytical queries run correctly and produce the expected results.

5. SQL Script and Final Submission – 1 Mark

The SQL script contains the table creation, data insertion, and analytical queries.

Total: 10 Marks

Evaluation Note

Marks should be awarded based on the work clearly visible in the submitted screenshots and SQL script.

If the required table, relationship, data, query, or result is not visible or cannot be verified, marks should not be awarded for that part.